

AGRONOMY

Undergraduate Study

Agronomy is the science and technology of producing plants that serve humans, using practices essential for maintaining and improving life. The Department of Agronomy offers a major leading to a degree of Bachelor of Science (B.S.) in agronomy. Graduates have the theoretical and practical knowledge needed for efficient and sustainable crop production. They are skilled in critical thinking, problem solving, communicating, and working effectively with others. They understand the ethical, cultural, and environmental dimensions of issues facing agronomists globally.

An agronomy major prepares students for employment in crop production and soil management, yield forecasting, precision farming, plant breeding, agricultural business and industry, agricultural service organizations, environmental and natural resource management, and farm management. Graduates pursue careers in the seed, fertilizer, and agricultural chemical industries as field agronomists, crop and soil management specialists, research technicians, sales and marketing specialists, and production managers. State and federal agencies employ agronomists as extension specialists, county extension directors, environmental and natural resource specialists, research associates, soil surveyors, soil conservationists, grain inspectors, integrated pest managers, land appraisers, agricultural lenders, and other science-based professionals.

An agronomy major also prepares students for graduate school. We offer a concurrent B.S./MBA degree. About a quarter of our students immediately continue into research-based M.S. and Ph.D. programs. As an undergraduate, there are many opportunities to be involved in research.

Department of Agronomy website - <http://www.agron.iastate.edu/>.
(<http://www.agron.iastate.edu/>)

Student Learning Outcomes

Upon graduation, students should be able to:

KNOWLEDGE: Graduates of our bachelor's degree program will demonstrate knowledge of the scientific principles and practices needed for success as an agronomist.

INTEGRATION AND PROBLEM-SOLVING: Graduates of our bachelor's degree program will integrate knowledge and skills to systematically assess, critically analyze, and sustainably manage agronomic systems.

PROFESSIONALISM AND CAREER READINESS: Graduates of our bachelor's degree program will demonstrate professional skills and career readiness.

COMMUNICATION: Graduates of our bachelor's degree program will communicate and engage effectively with diverse audiences.

DIVERSE WORLDVIEWS: Graduates of our bachelor's degree program will demonstrate knowledge of diverse worldviews that affect agronomic practices.

ETHICAL, ENVIRONMENTAL, AND ECOLOGICAL ASPECTS: Graduates of our bachelor's degree program will demonstrate knowledge of the ethical, environmental, and ecological aspects of agronomic practices.

Curriculum in Agronomy

Total Degree Requirement: 120 cr.

Only 65 cr. from a two-year institution may apply which may include up to 16 technical cr.; 9 P-NP cr. of free electives; 2.00 minimum GPA. A minimum of 15 credits of agronomy courses must be earned at Iowa State and not transferred from other institutions.

International Perspective: 3 cr.

3 cr. from approved International Perspective list: <http://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current> (<http://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current/>)

U.S. Cultures and Communities: 3 cr.

3 cr. from approved U.S. Cultures and Communities list: <http://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses> (<https://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses/>)

Communication/Library: 13 cr.

6 cr. of English composition with a C or better and 3 cr. of speech fundamentals with a C or better.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
SPCM 2120	Fundamentals of Public Speaking	3
	or AGEDS 3110 Presentation and Sales Strategies for Agricultural Audiences	
LIB 1600	Introduction to College Level Research	1
One of the following:		
ENGL 3020	Business Communication	3
ENGL 3090	Proposal and Report Writing	3
ENGL 3120	Communicating Science and Public Engagement	3
ENGL 3140	Technical Communication	3

Humanities: 3 cr.

3 cr. from approved humanities list: <http://www.cals.iastate.edu/student-services/humanities> (<http://www.cals.iastate.edu/student-services/humanities/>)

Social Sciences: 3 cr.

3 cr. from approved social sciences list: <http://www.cals.iastate.edu/student-services/social-sciences> (<http://www.cals.iastate.edu/student-services/social-sciences/>)

Ethics: 3 cr.

3 cr. from approved ethics list: <http://www.cals.iastate.edu/student-services/ethics> (<http://www.cals.iastate.edu/student-services/ethics/>)

Mathematical Sciences: 6 cr.

MATH 1400	College Algebra	3
STAT 1040	Introduction to Statistics	3

Physical Sciences: 8 cr.

CHEM 1630 & 1630L	College Chemistry and Laboratory in College Chemistry	5
One of the following:		
AGRON 2590	Organic Compounds in Plants and Soils	3
BBMB 2210	Structure and Reactions in Biochemical Processes	3
CHEM 2310 & 2310L	Elementary Organic Chemistry and Laboratory in Elementary Organic Chemistry	4

Life and Biological Sciences: 6 cr.

BIOL 2510	Biological Processes in the Environment	3
AGRON 3200 or BIOL 3130	Genetics, Agriculture and Biotechnology Principles of Genetics	3

Supporting Sciences: 12 cr.

Courses cannot be used to fulfill any other university, college, or Agronomy requirements. At least 9 cr. must be in courses numbered 3000 or above.

This requirement can be met in one of two ways:

a. Complete 3 cr. in the following: (ACCT, BBMB, BIOL, CHEM, COMS, DS, ECON, All ENG, GEOL, GEN, MATH, MTEOR, PHYS, STAT) as well as 9 additional cr. in the following: (ANS, BIOL, BBMB, DPA, ENSCI, ENT, GEOL, HORT, GEN, MICRO, NREM, PLP, TSM and/or AGRON). At least 9 cr. must be in courses numbered 3000 or above.

b. Complete the courses needed to fulfill a second major that complements the student's academic and professional goals.

Agronomy Core: 45 cr.

AGRON 1050	Leadership Experience	R
AGRON 1100	Professional Development in Agronomy: Orientation	1
AGRON 1800	Global Agriculture in a Changing World	3
AGRON 1810	Introduction to Crop Science	3
AGRON 1820	Introduction to Soil Science	3
AGRON 1830	Basic Skills for Agronomists	1

AGRON 2060	Introduction to Weather and Climate	3
AGRON 2100	Professional Development in Agronomy: Career Planning	1
AGRON 2790	Field Exploration of Agronomy	3
AGRON 2800	Crop Development, Production and Management	3
AGRON 2820	Soil Conservation and Land Use	3
AGRON 3100 or AGRON 311C	Professional Development in Agronomy: Work Experience Professional Internship in Agronomy	R
AGRON 3160	Crop Structure-Function Relationships	3
AGRON 3420 or AGRON 450C	World Food Issues: Past and Present Issues in Sustainable Agriculture	3
AGRON 3540 & 3540L	Soils and Plant Growth and Soils and Plant Growth Laboratory	4
AGRON 3830	Evaluation of Agronomic Systems	4
AGRON 4100	Professional Development in Agronomy: Senior Forum	1
Additional AGRON credits at the 3000-4000 level		6

Electives: 15 cr.

Additional free electives 15

Agronomy, B.S.

Freshman

Fall	Credits Spring	Credits
AGRON 1100	1 AGRON 1810	3
AGRON 1800 (Intl Persp)	3 AGRON 1820	3
AGRON 1830	1 AGRON 2100	1
BIOL 2510	3 CHEM 1630 or 1770	4
ENGL 1500	3 CHEM 1630L or 1770L	1
LIB 1600	1 ENGL 2500	3
MATH 1400	3	
	15	15

Sophomore

Fall	Credits Spring	Credits
AGRON 2060	3 AGRON 2820	3
AGRON 2790	3 SPCM 2120 or AGEDS 3110 (Speech choice)	3
AGRON 2800	3 Organic Chemistry: AGRON 2590, BBMB 2210, or CHEM 2310 & L	3-4
STAT 1040	3 Elective	3
Humanities	3 Social Sciences	3
	15	15-16

Junior			AGRON 1800	Global Agriculture in a Changing World	3	
Fall	Credits	Spring	Credits	AGRON 2060	Introduction to Weather and Climate	3
AGRON 3200 or BIOL 3130 (Genetics Choice)	3	AGRON 3160	3	AGRON 2800	Crop Development, Production and Management	3
AGRON 3540	3	AGRON 3420 or 4500 (Ethics)	3	AGRON 2820	Soil Conservation and Land Use	3
AGRON 3540L	1	AGRON 3830	4	Supporting Courses (6-9 credits, 6 credits of which must be 3000+ level):		
Elective	6	ENGL 3020, 3090, 3120, or 3140	3	AGRON 2170	Weed Identification	1
Supporting Sciences	3	Elective	3	AGRON 2590	Organic Compounds in Plants and Soils	3
	16		16	AGRON 2700	Geospatial Technologies	3
Senior			AGRON 3160	Crop Structure-Function Relationships	3	
Fall	Credits	Spring	Credits	AGRON 3170	Principles of Weed Science	3
Additional Agronomy credits (3000+)	3	AGRON 4100	1	AGRON 3200	Genetics, Agriculture and Biotechnology	3
Elective	6	Additional Agronomy credits (3000+)	3	AGRON 3340	Forage Crop Management	3
Supporting Sciences (3000+)	3	Elective	3	AGRON 3380	Seed Science and Technology	3
U.S. Cultures and Communities	3	Supporting Sciences (3000+)	6	AGRON 3540	Soils and Plant Growth	3
	15		13	AGRON 3600	Environmental Soil Science	3
Minor - Agronomy			AGRON 3920	Systems Analysis in Crop and Soil Management	3	
A minor in agronomy is earned by taking 18 credits consisting of the following: • 6 credits of required courses; • 3-6 credits of foundation courses;			AGRON 4050	Soil-Plant-Animal-Atmosphere Physics	3	
			AGRON 4210	Introduction to Plant Breeding	3	
			AGRON 4500	Issues in Sustainable Agriculture	3	
			AGRON 4590	Soil and Water Chemistry	3	
			AGRON 4590L	Soil and Water Chemistry Laboratory	1	
			AGRON 4630	Soil Formation and Landscape Relationships	3	
			AGRON 4770	Soil Physics	3	
			AGRON 4840	Organic Agricultural Theory and Practice	3	
			AGRON 4850	Soil and Environmental Microbiology	3	
			AGRON 4970	Agroecology Field Course	3	

Minor - Agronomy

A minor in agronomy is earned by taking 18 credits consisting of the following:

- 6 credits of required courses;
- 3-6 credits of foundation courses;
- 6-9 credits of supporting courses.

At least 6 credits numbered 3000 or above are required and at least 3 credits must be taken at Iowa State University. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement.

A student may use up to:

- 3 credits of AGRON 4960A or AGRON 4960B;
- **Or** up to 2 credits from AGRON 3310 or AGRON 3700;
- **Or** up to 1 credit of AGRON 4900E or AGRON 4900G or AGRON 4900H or AGRON 4900Z

Required Courses (6 credits):

AGRON 1810	Introduction to Crop Science	3
AGRON 1820	Introduction to Soil Science	3

Foundation Courses - One or two courses from the list below (3-6 credits):

SOIL SCIENCE CERTIFICATE

Purpose

The certificate in soil science will prepare students with the skills needed for successful careers in the field of soil science. Soil science has key roles in achieving goals of sustainability as demonstrated by the functions of soil and their ecosystem services. Namely, soil is an important factor in addressing issues of food scarcity, water management, climate change, biodiversity loss, and human health. Students who complete the Iowa State University soil science certificate program will qualify for federal employment as a soil scientist. In addition, students completing the certificate will have met the educational component of the Soil Science Society of America's requirements for certification as a professional soil scientist and all U.S. states' educational criteria for licensure as a professional soil scientist.

Requirements

The certificate in soil science requires the completion of 30 credit hours, including a foundation course in soil science, 15 credits in supporting biological, physical, or Earth sciences, as well as 12 credits from approved categories of soil science courses.

At least 9 credits must be taken in soil science courses numbered at the 3000-level or above.

Courses for the certificate cannot be taken on a pass/not-pass basis.

Course Requirements for Soil Science Certificate

Foundation: 3 credits of required core course.

AGRON 1820 Introduction to Soil Science 3

Supporting Biological, Physical, or Earth Sciences: Choose 15 credits from approved list consisting of courses from the following designators: ABE, AGRON, AECL, BIOL, CHEM, CE, CRP, ENSCI, ENT, FOR, GEOL, HORT, LA, MTEOR, MICRO, NREM, PHYS.

Soil Science: Choose 12 credits with at least 2 credits from each of the following categories.

Soil Physical Properties or Soil Water Relationships (2 credits)

AGRON 2820	Soil Conservation and Land Use	3
AGRON 3600	Environmental Soil Science	3
AGRON 4050	Soil-Plant-Animal-Atmosphere Physics	3
AGRON 4770	Soil Physics	3
ABE 4310	Design and Evaluation of Soil and Water Conservation Systems	3
TSM 3240	Soil and Water Conservation Management	3

Soil Chemistry (2 credits)

AGRON 2590	Organic Compounds in Plants and Soils	3
AGRON 4590	Soil and Water Chemistry	3
AGRON 4590L	Soil and Water Chemistry Laboratory	1

Soil Biology (2 credits)

AGRON 3540	Soils and Plant Growth	3
AGRON 3540L	Soils and Plant Growth Laboratory	1
AGRON 4850	Soil and Environmental Microbiology	3

Soil Morphology and Geography (2 credits)

AGRON 2500	Environmental Geography	3
AGRON 2700	Geospatial Technologies	3
AGRON 3700	Field Experience in Soil Description and Interpretation	1

AGRON 4630	Soil Formation and Landscape Relationships	3
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The Soil Science Certificate may be earned at the same time as an ISU baccalaureate degree or after the completion of an associate of arts (AA)

or associate of science (AS) degree or a baccalaureate degree from any accredited institution.

Information on the concurrent Agronomy bachelor's degree and MBA can be located here (<https://www.ivybusiness.iastate.edu/full-time-concurrent-mba/>).

Graduate Study

The Department of Agronomy seeks and promotes ways to manage and protect the resources that generate the world's agricultural productivity and wealth. The department offers programs leading to Master of Science (MS) and Doctor of Philosophy (PhD) degrees with majors in agricultural meteorology, crop production and physiology, plant breeding, and soil science. Online MS programs in agronomy and plant breeding are available, which are suitable for professionals working in industry or government. The MS in agronomy program also offers a graduate certificate. A graduate minor is offered in all these areas to on-campus students. Candidates pursuing the MS degree may do so by completing either a thesis or nonthesis academic plan. The department also cooperates with the interdepartmental majors in environmental science, genetics and genomics, plant biology, and sustainable agriculture.

A strong undergraduate program with fundamental training in biology, chemistry, mathematics, physics, and statistics is required for students interested in graduate study. Academic Plans are designed individually per the requirements of the Graduate College and each departmental major. Graduate students are mentored by their major professor and a committee of supporting faculty.

Our on-campus graduate programs require students to be sponsored by a faculty member before admission. Assistantships are largely project-based, with the goal of participating in independent research to complete a thesis or dissertation.

The specific requirements of each graduate program can be found in its program handbook, which is available through the Graduate College.